From 15,099 player-seasons to a data-driven shortlist

All filtering is automated using data within our own dataset. Player ages are calculated from birth_date in player_info.csv — no external research required to remove veterans.

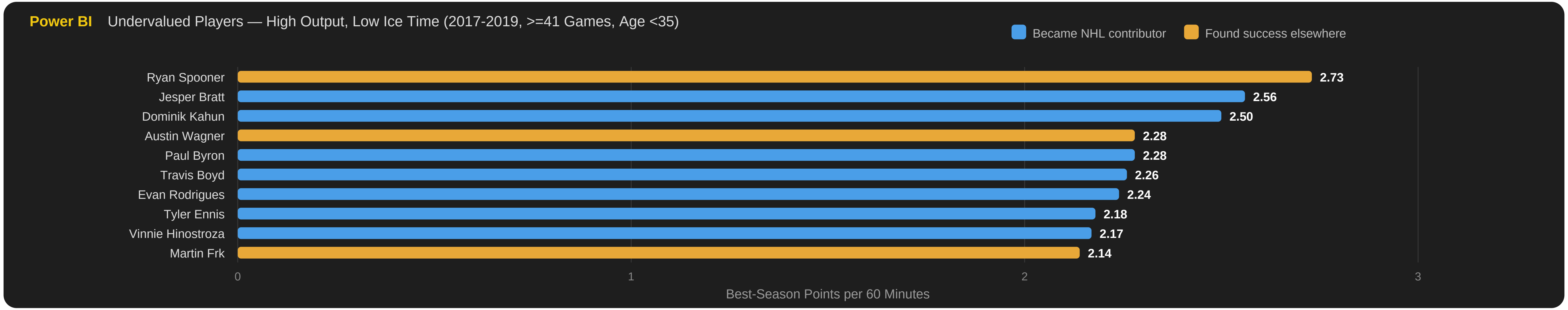
1.57 pts/60
League avg efficiency threshold (2017-2019)

33.74 hrs
League avg TOI/season threshold (2017-2019)

Age < 35
Veteran cutoff from birth_date in dataset

10 players
Shortlist after all automated filters

POWER BI DASHBOARD · mart_player_season · pts/60 > 1.57 avg · TOI < 33.74 hrs avg · >=41 games · 2017-2019 · is_veteran = False (age < 35)



How is_underused_high_impact is calculated

pts/60 > 1.57 (league avg) AND total season TOI < 33.74 hrs (league avg). Both thresholds recalculate each season dynamically.

How is_veteran is calculated

Age from birth_date in player_info.csv: FLOOR(DATEDIFF(days, birth_date, 2019-12-31) / 365.25) >= 35. Removes veterans automatically — no external research needed.

10 names on the shortlist — what happened after 2019?

Ranked by the pipeline (best-season pts/60). All filters automated from within the dataset; ages from player_info.csv remove veterans. Dataset ends 2019 — here is what happened next.

REAL-WORLD OUTCOMES AFTER 2019 · Ranked by pipeline pts/60 · Verified June 2026

#1 **Ryan Spooner** — **2.73 pts/60**
Age 27 at end of dataset
Left NHL after 2019 but became a KHL star — 109 pts in 142 games with Avangard. Still playing.
Found success elsewhere

#2 **Jesper Bratt** — **2.56 pts/60**
Age 21 at end of dataset
New Jersey Devils franchise cornerstone & alternate captain. Five straight 70+ point seasons.
Franchise cornerstone

#3 **Dominik Kahun** — **2.50 pts/60**
Age 24 at end of dataset
83 NHL points across Chicago, Pittsburgh, Buffalo & Edmonton. IIHF silver 2023. Now Lausanne HC.
Multi-team contributor

#4 **Austin Wagner** — **2.28 pts/60**
Age 22 at end of dataset
Reliable bottom-6 forward, 178 NHL games. Now in the KHL (Shanghai Dragons).
Found success elsewhere

#5 **Paul Byron** — **2.28 pts/60**
Age 30 at end of dataset
Montreal alternate captain. Played all 22 games of the Canadiens 2021 Stanley Cup Final run.
Cup finalist & leader

#6 **Travis Boyd** — **2.26 pts/60**
Age 26 at end of dataset
Two 30+ point seasons at Arizona (career-high 35) — double the minutes Washington gave him.
Breakout elsewhere

#7 **Evan Rodrigues** — **2.24 pts/60**
Age 26 at end of dataset
Won back-to-back Stanley Cups with Florida Panthers (2024 & 2025). 4G/3A in the 2024 Final.
Two-time Cup champion

#8 **Tyler Ennis** — **2.18 pts/60**
Age 30 at end of dataset
Suppressed at Toronto once Nylander returned. 37 points next season at Ottawa & Edmonton given minutes.
Breakout elsewhere

#9 **Vinnie Hinostroza** — **2.17 pts/60**
Age 25 at end of dataset
Veteran journeyman, 460+ NHL games. Still in the NHL system 2025-26 (Minnesota, then Florida).
Still in NHL system

#10 **Martin Frk** — **2.14 pts/60**
Age 26 at end of dataset
Elite AHL scorer — 200+ goals & the hardest shot in AHL history. Offense-first game; brief NHL stints.
Found success elsewhere

All 10 shortlisted players went on to meaningful pro careers

7 became established NHL contributors; the other 3 found success elsewhere (Spooner & Wagner in the KHL, Frk an elite AHL scorer). Evan Rodrigues won back-to-back Cups with Florida; Travis Boyd and Tyler Ennis broke out once given real ice time. Not one shortlisted player washed out of pro hockey — the pipeline surfaced genuine talent.

Where do shots actually become goals?

The slot converts at 19.46% — six times higher than wing shots. Practice time should follow the data, not tradition.

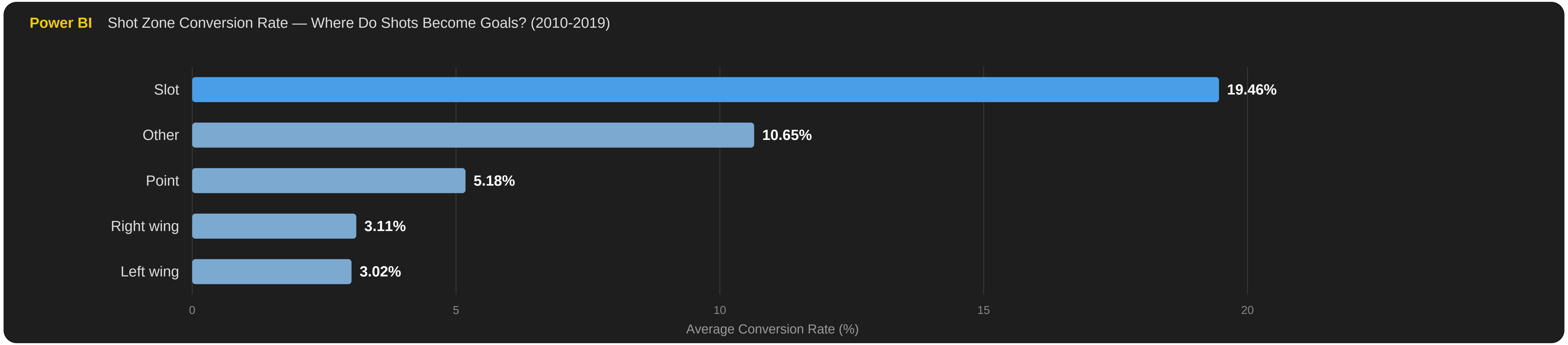
19.46%
Slot conversion rate

6x
More effective than the wings

9 vs 33
Slot vs wing shots per goal

1.7M+
Shot events analysed (2010-19)

POWER BI DASHBOARD · mart_shot_zones · conversion rate by zone, 2010-2019



The finding

Slot shots convert at 19.46% vs ~3% from the wings. The hierarchy holds across every season 2010-2019. Pre-2010 excluded (100% missing coordinates).

The recommendation

Drill slot positioning and finishing, not perimeter technique. The problem is rarely the shot — it is where the player is standing when they take it.

CALL 3 — THE PENALTY

When do penalties actually cost games?

A Period 3 Game Misconduct correlates with losing two times out of three. A Period 1 Minor is a coin flip. Severity and timing both matter.

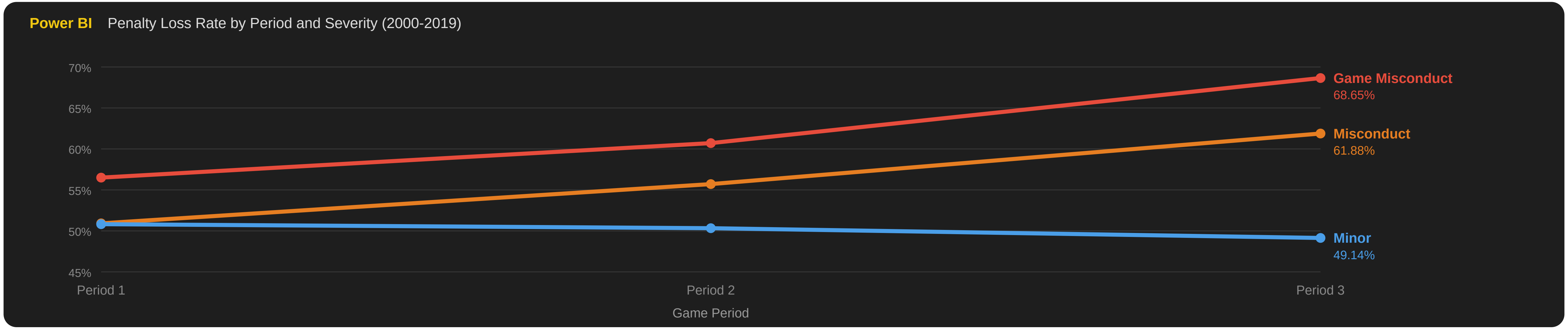
68.65%
Period 3 Game Misconduct loss rate

50.82%
Period 1 Minor — a coin flip

12pp
P3 vs P1 gap, same penalty type

247K+
Penalty records analysed

POWER BI DASHBOARD · mart_penalty_cost · loss rate by period and severity, 2000-2019



Correlation, not causation

Teams already losing may take more desperate penalties. But the same penalty type carries a 12-point higher loss rate in Period 3 than Period 1 — if score state alone explained it, that gap would be far smaller.

The recommendation

Coach discipline differently by period. A Period 1 Minor is noise. A Period 3 Game Misconduct costs the game two times out of three. The pre-Period-3 locker room message should be specifically about composure.

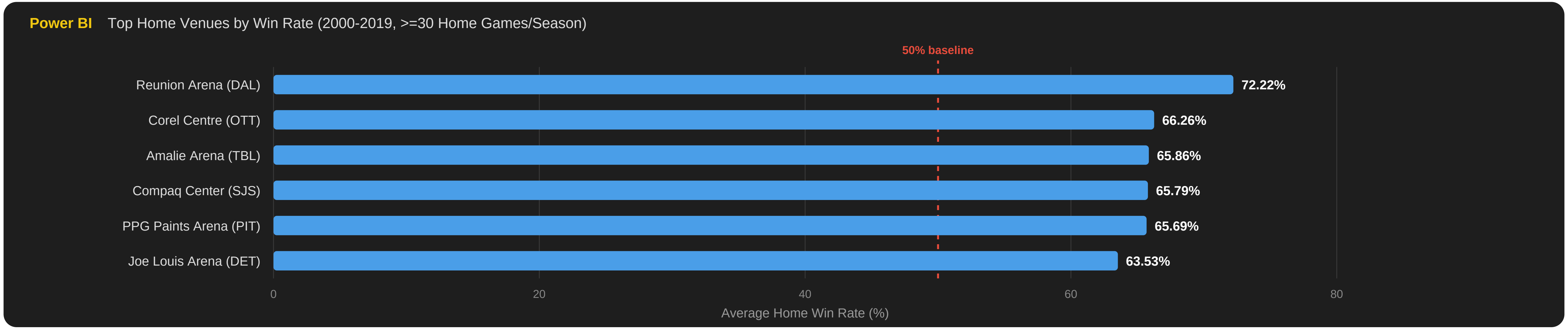
Pipeline: Polars · DuckDB · dbt-core · Power BI · GitHub Actions · Verified June 2026

How much does home ice actually matter?

Home ice advantage is worth +13 to +22 percentage points in win probability at top NHL venues — and nearly a full extra goal per game.

+13-22pp	**+0.9**	**65.86%**	**All teams**
Win rate uplift at top home venues	More goals per game at home	Tampa Bay Lightning at Amalie Arena	Every venue between 55-72% home wins

POWER BI DASHBOARD · mart_venue_advtg · top home venues by win rate, >=30 home games/season, 2000-2019



The finding

Every qualifying venue wins between 55% and 72% at home — home advantage is league-wide, not arena-specific. Home teams score ~0.9 more goals per game. (Reunion Arena is a small-sample outlier.)

The recommendation

Finishing top of the division to earn home playoff games is worth more than most trade deadline acquisitions. In a Game 7, home ice is the difference between a coin flip and winning two times out of three.

Which franchises are rising or falling?

A 3-year rolling win rate classifies every franchise by consistent directional momentum as of 2019 — before the standings confirmed it.

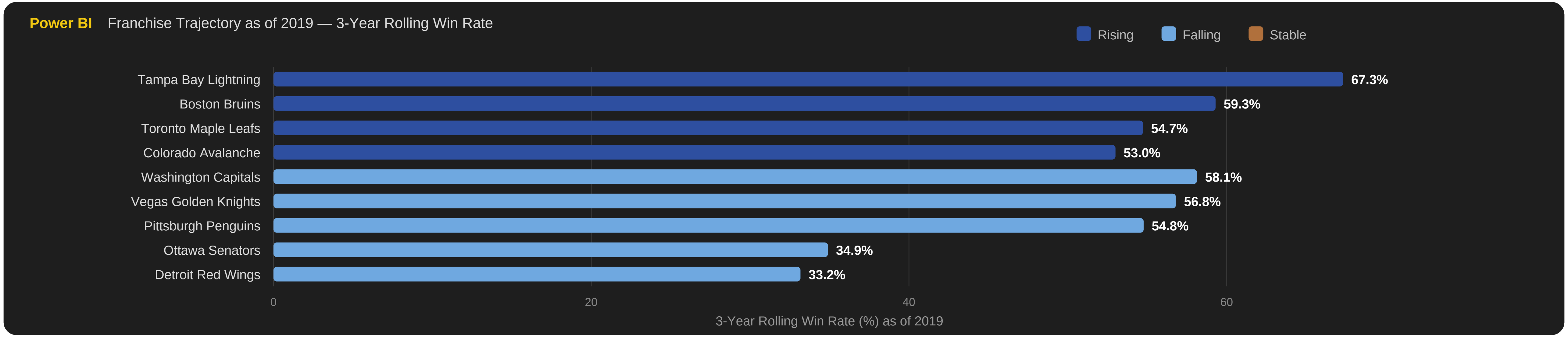
9 / 12 / 12
Rising / stable / falling franchises

+11.17pp
Colorado Avalanche — largest rise

-10.07pp
Ottawa Senators — steepest fall

momentum
Consistent 3-season direction, not net change

POWER BI DASHBOARD · mart_team_traject_rolling · 3-year rolling win rate as of 2019, colour-coded by trajectory



Trajectory measures consistent momentum, not net change

A small but unbroken decline counts as falling; a larger but inconsistent swing counts as stable. This page shows the top rising and falling franchises as of 2019. See page 2 for what happened next.

Our dataset ends in 2019 — what happened next?

The pipeline classified franchises in 2019. We checked the real-world outcomes through 2026. A compass, not a crystal ball.

RISING FRANCHISES — VERIFIED OUTCOMES AFTER 2019

Tampa Bay Lightning +9.16pp
Won back-to-back Stanley Cups in 2020 and 2021.
Champion

Colorado Avalanche +11.17pp
Won the Stanley Cup in 2022 — largest upward momentum of any team.
Champion

Boston Bruins +5.30pp
Set the all-time NHL regular season wins record in 2022-23 (65 wins).
Record-setter

Carolina Hurricanes +7.97pp
Seven consecutive playoff appearances through 2025.
Contender

Toronto Maple Leafs +7.56pp
Eight straight playoffs; first series win since 2004 in 2023.
Contender

New York Islanders +2.30pp
Back-to-back Conference Finals in 2020 and 2021.
Contender

FALLING FRANCHISES — VERIFIED OUTCOMES AFTER 2019

Detroit Red Wings -8.53pp
10 consecutive seasons without playoffs — longest active drought.
Validated

Washington Capitals -6.07pp
Have not won since 2018; now in rebuild phase.
Validated

Pittsburgh Penguins -4.73pp
Have not won since 2017; entering rebuild.
Validated

Ottawa Senators -10.07pp
Years rebuilding; returned to playoffs only in 2024-25.
Validated

Vegas Golden Knights -5.90pp
Won the Stanley Cup in 2023 — fell from expansion peak then rebuilt.
Miss

New York Rangers -7.07pp
Rebounded to reach the Conference Finals by 2024.
Miss

In 2019, this pipeline already knew who would lift the Cup

Tampa Bay Lightning and Colorado Avalanche — both flagged rising — won within three years. Two misses (Vegas Golden Knights, New York Rangers) show the metric identifies trends, not certainties. The pipeline is a compass, not a crystal ball.